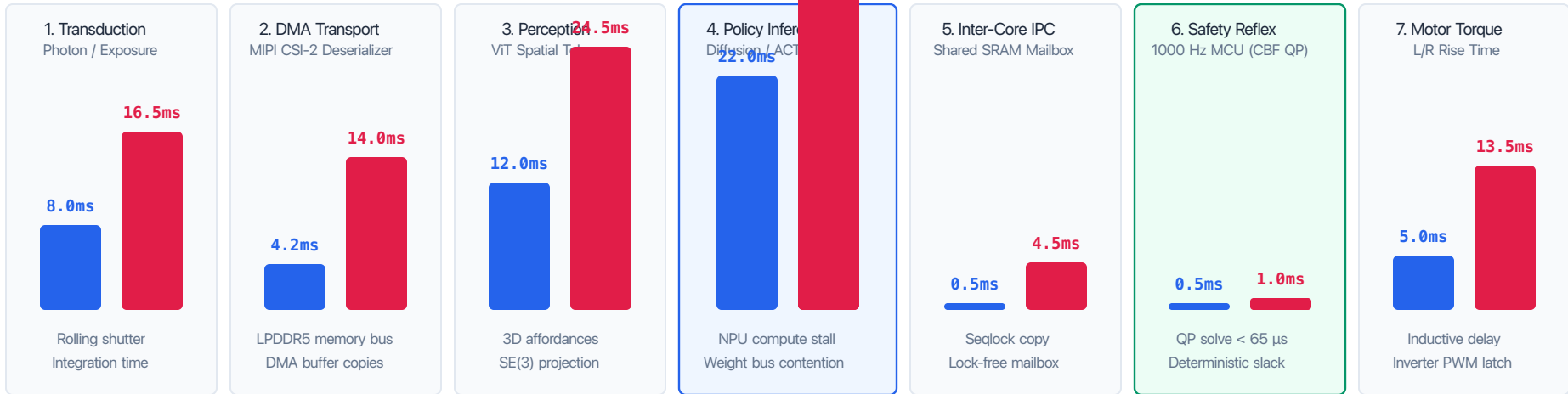


# THE 7-STAGE SENSE-TO-ACTUATION LATENCY WATERFALL

Deconstructing where time is lost: Nominal P50 (52.2 ms) vs. P99 Latency Tail (126.0 ms) Breaching World Deadlines



■ P50 Nominal Execution

■ P99 Contention & Jitter Tail Blowout

----- Physical World Stability Deadline ( $\tau_{\text{world}} = 100 \text{ ms}$ )

P50 Total: 52.2 ms (Safe Closed-Loop Margin: +47.8 ms)

P99 Total: 126.0 ms (BREACH!)

$\tau_{\text{world}} = 100 \text{ ms}$

**Systems Insight:** While mean latency ( $P50 = 52.2 \text{ ms}$ ) looks safe in benchmarking, real-world memory bus contention and OS scheduling inflate  $P99$  past the world deadline, resulting in open-loop drift and physical